Ariana Feng

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EDUCATION

Western University — London, ON Sep 2022 – Apr 2027 (Expected)

Honours B.Sc. Applied Mathematics | B.Sc. Computer Science

- Western Science Co-op Program
- Western Scholarship of Distinction

TECHNICAL SKILLS

Languages: Python, C++, C#, SQL, JavaScript

Frameworks & Libraries: React, Node.js, ASP.NET MVC, Qt6, Hugging Face, XGBoost, SHAP, GoogleTest

Tools & Platforms: Git, SQL Server, Active Directory, LDAP, SSO, ServiceNow, IIS, F5

Concepts: Machine Learning, Statistical Modeling, Relational Database Design, RBAC, Semantic Retrieval, REST/API Integration

WORK EXPERIENCE

Software Developer, Core Platform Services May 2026 – Sep 2026

Royal Bank of Canada — Toronto, ON

- Designed and built an enterprise Accomplishment Portal from scratch (React, Node.js, Python, SQL Server), centralizing accomplishment reporting across multiple organizational groups.
- Designed and iteratively refined relational database schemas to improve data organization, scalability, and maintainability.
- Implemented authentication and authorization using LDAP, SSO, and Active Directory group-based RBAC for managers, approvers, and executives.
- Developed role-specific dashboards, approval workflows, audit history, and validation pipelines to improve reporting consistency and traceability.
- Worked in Kanban with feature branching, pull requests, and code reviews; presented architecture, progress, and live demos to Vice Presidents and senior leadership.

Technical Systems Analyst, Web Infrastructure May 2025 – Dec 2025

Royal Bank of Canada — Toronto, ON

- Built internal full-stack web applications (ASP.NET MVC, C#) to automate infrastructure tasks for the Windows middleware team, including application pool operations, NAS permission visibility, SSL certificate monitoring, and change record validation.
- Integrated tools with Active Directory, ServiceNow, Chorus API, IIS Administration API, and F5.
- Designed and built WebinfraAI, an AI-assisted ticket resolution system using semantic retrieval, intent classification, and Hugging Face language models with a custom classifier to improve routing of structured and unstructured support requests.
- Built retrieval pipelines matching infrastructure queries to historical configuration records and technical documentation.

PROJECTS

Market-Making Simulator & Optimal Quoting | C++, Qt6, GoogleTest Sep 2026 – Dec 2026 (Expected)

- Developing an extensible market-making simulator in C++ modeling limit order books, stochastic order flow, and inventory risk.
- Implementing two quoting modes: manual quoting and selectable automated strategies, starting with a fixed-spread strategy and an inventory-skew strategy that adjusts quotes based on current position.

Undergraduate Thesis: ODE Inverse Problems using Machine Learning Sep 2026 – Present

- Research thesis applying machine learning to inverse problems for ordinary differential equations, supervised by Prof. Agustinus Kristiadi (Western University).

Household Savings & Investment Behavior Analysis | Elastic Net, XGBoost, SHAP 2024

- Analyzed large-scale household financial datasets to study savings and investment behavior across demographic groups.
- Applied clustering and predictive models (Elastic Net, XGBoost) to identify financial behavior patterns and estimate savings contribution.
- Used SHAP interpretability analysis to identify the key factors driving long-term investment activity.